# 21

# Virtual LAN

*Virtual LAN or VLAN as it is called, is primarily a performance enhancement method and aims at limiting broadcast traffic being limited to segments to which they are meant. This also results in improved security though security is not the primary purpose of a VLAN implementation. VLAN cannot be the sole method of Network security but has to be implemented in combination with other methods discussed in the previous modules.*

## Objectives

On completing the study of this module, you will learn about:

- Need for Virtual LAN (VLAN), benefits and constraints
- Operating principle of a VLAN
- Methods of implementing a VLAN
- Possible methods of connection
- Filtering table
- Tagging

## 21.1 Introduction

We have learnt in earlier modules about *Local Area Networks* (LANs) and about the different types of networks such as Ethernet, Token Ring and FDDI. We have also had a look at various network connectivity devices such as repeaters and hubs. Ethernet is by far the most popular networking topology adopted the world over for interconnecting computers. Switched Ethernet has become almost a *de facto* standard with the widespread availability of rugged and low cost switches. Since Ethernet uses a collision sensing type of communication protocol, large networks face problems of network throughput. Segmenting the network somewhat mitigates this problem.

When a LAN is divided into segments using a switch, with each port serving a smaller number of network nodes, the chances of collision reduce. Moreover, the devices that normally communicate with one another are placed in one segment so that the need for forwarding the packets to other ports also gets reduced. In some cases, machines that

require very high bandwidth (for example, a server or a high performance workstation) are connected directly to a switch port, thus enabling them to have almost the entire bandwidth of one segment dedicated to them.

We will learn in this chapter how Virtual LAN (VLAN) technology improves the LAN performance and also helps in making the network more secure. At the same time, it should be noted that security is not the main consideration in setting up VLANs nor they can be the sole approach to network security. When used in combination with other security implementations, a VLAN can result in improved network security.

## 21.2 Need for VLAN

To begin with, LANs were just that: networks that served users in a small area, usually a small department or a workgroup with one server and a few clients. Very often, different floors in an office each had a LAN. Sometimes these LANs were connected by routers to obtain a set of interconnected LANs (Figure 21.1).

The routers, which connected the LANs, enabled communication between LANs but they were often complicated, slow and expensive. With the advent of Ethernet switches a need for a better solution was felt. Routers were therefore pushed to the periphery, say, for example, communication links leading to the external world (a WAN or the Internet). See Figure 21.2.

A switched LAN also has some disadvantages. While switches are able to segment unicast traffic (one node to another), multicast or broadcast traffic is allowed to pass through, unlike routers. In other words, while each segment is a collision domain, the entire LAN is a broadcast domain. This can become a bottleneck restricting LAN throughput.

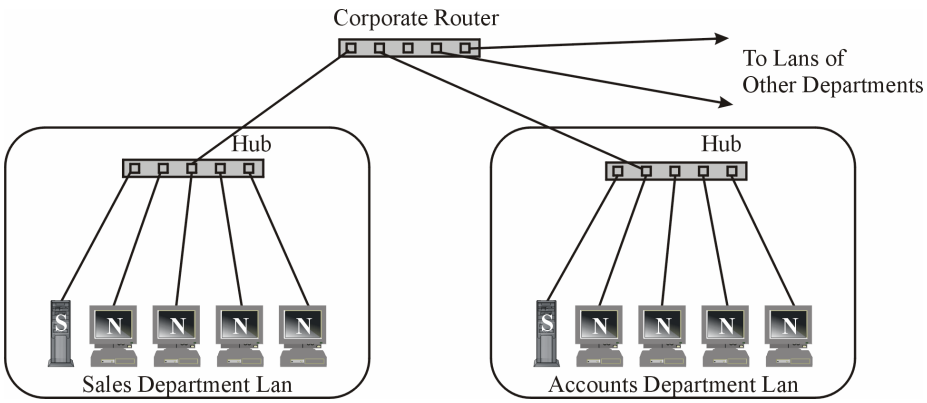



**Figure 21.1**
*Interconnected LANs*

The other factor, which affects performance, is due to changes in the present business environment. Very often the personnel involved in a particular project or those belonging to a particular department are not confined to a given area and are spread throughout a building or campus. Product design teams may be cross functional groups and usually exist for short periods of time. In such cases, grouping the users into one physical segment is not feasible. Refer to figure 21.3. In these cases, more packets have to travel from one physical segment (or switch port) to another, thus increasing the network loading. VLANs offer a way to overcome these problems.

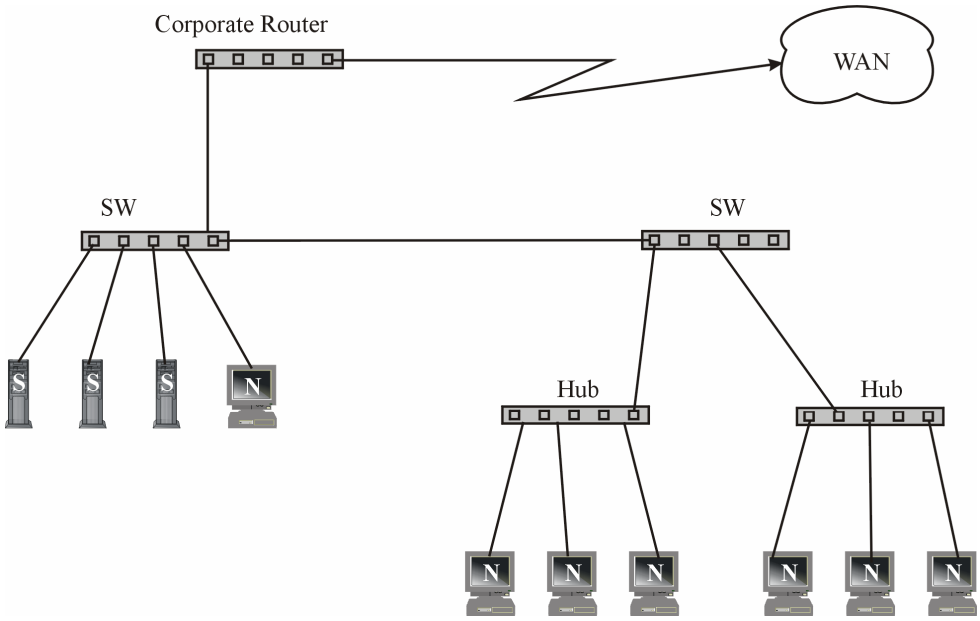



**Figure 21.2**
*A switched LAN*

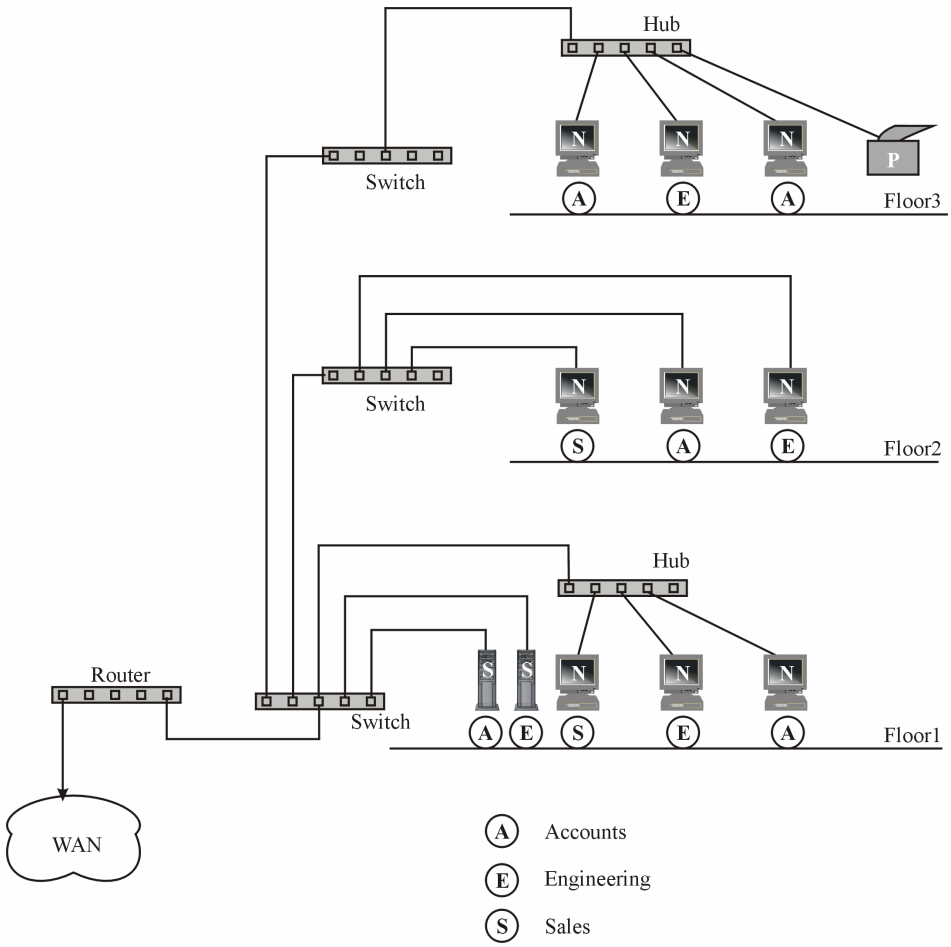



**Figure 21.3**
*Workgroup distribution within a building*

A VLAN logically groups switch ports into workgroups. Since broadcasts and multicasts between the users of a workgroup are likely to be high, a VLAN that includes members of a given workgroup limits the broadcast traffic to within the particular virtual network. Thus a VLAN performs like a virtual broadcast domain. Figure 21.4 shows the logically defined VLAN network.

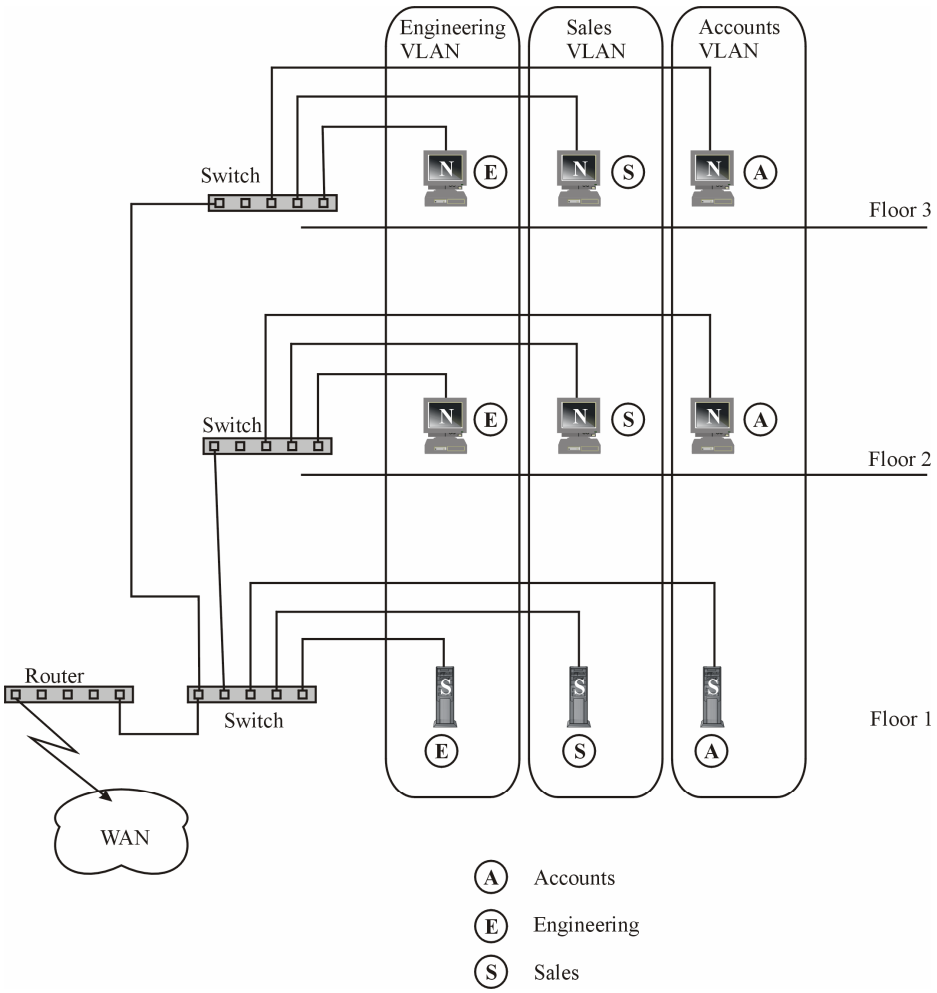



**Figure 21.4**
*A typical network of VLANs*

## 21.3 Benefits of a VLAN

VLANs offer a number of advantages over the traditional LAN implementation. We will discuss these advantages below:

### Performance improvement

In networks having a high proportion of broadcast traffic, a VLAN can improve network performance by limiting the broadcast from going to destinations for which the broadcast is not intended. No doubt a router could perform this function, but the greater amount of processing required for a router to route the frames or packets increases latency and therefore reduces the performance of the network. VLAN is simply more efficient.

### Improved security

By the ability to contain the broadcast activity within a workgroup, access to sensitive data broadcast on a network is limited only to those for whom such broadcast is intended. This improves data security of the network.

### Ability to set up virtual workgroups

As discussed earlier in this chapter, VLAN makes it possible for the network administrator to define a set of users as belonging to a virtual broadcast domain irrespective of their physical location. The only alternative to VLAN is to physically move them. Given that present day organizations are more dynamic, frequent changes in workgroups have also become necessary. This can be achieved by VLAN without the need of physical relocation every time a change takes place.

### Reduced administration

A considerable part of the administration efforts go towards additions, movements, and changes, all of which involve reconfiguration of hubs, routers, and station addressing and sometimes re-cabling. A rough estimate is that 70% or more of administration time is spent on such activities. A VLAN reduces the need for such changes and with good management tools; all that has to be done is a simple drag and drop action to change a user from one VLAN to another.

### Reduced cost

The ease of administration, the avoidance of physical movement/cabling changes and elimination of the need for expensive routers to contain broadcast activity keep the costs lower.

## 21.4 VLAN constraints

VLANs are not without problems. When resources such as a printer have to be shared within a logical group, it may cause problems for some who will have the printers assigned to them on another floor of the office far from where they work.

Also, the present trend is to group different servers in a common server farm, with increased physical security, environment control, fire prevention measures etc. These servers may have to be accessed by members of more than one VLAN. If it is not possible to assign the server to more than one VLAN, such access is not possible, unless the servers are put in a separate VLAN of their own and connected to the rest of the network by a router. This can slow down network throughput.

Thus the ability for a node to be assigned to more than one VLAN is a critical success factor in a VLAN implementation. The constraints pointed out in the previous factor cannot be otherwise overcome.

Also, VLAN implementations are by and large proprietary, which makes inter-operation between switches difficult. This means that anyone wanting to establish a VLAN has to procure all products from the same vendor. Standards for VLAN such as IEEE 802.1Q are still under evolution and in the absence of such standardization VLANs remain somewhat restrictive.

## 21.5 Operating principle of a VLAN

We will see how a VLAN functions as a virtual broadcast domain. The switch is the core component of any VLAN implementation. It serves as the point of communication to data destined for a node or data received from a node connected to it. The network

administrator decides how VLANs will be grouped. The switch implements this decision. The switch provides the intelligence required to identify the group to which a packet belongs. This can be carried out either by packet filtering or by packet identification (also called tagging).

Packet filtering is similar to the technique used by routers. In this method, each packet received by the switch is examined and compared with a filtering table (also called a filtering database). The filtering table that is developed for each switch contains information on user groups based either on port address, MAC (Media Access Control) station address, protocol type or application type (we will learn about these in greater detail later in this chapter). The switch takes appropriate action based on the comparison. Packet filtering thus provides an additional layer of processing for deciding as to how the packet is to be handled. It thus increases the switch latency. It is also necessary that all the switches in the network must maintain and synchronize their filtering tables, which involves administrative overheads.

Packet identification is a relatively new method and consists of placing a unique identifier in the header of each packet as it travels through the switch fabric. The identifier is examined by each switch prior to any broadcast or transmissions to other switches/routers or workstations in the network. When a packet exits the switch fabric (routed to the target end station) the switch removes the identifier from the header. Packet identification functions in layer 2 and does not involve administrative overheads.

The latter method of adding an identifier to the header for identification is called explicit tagging. On the other hand, the method of packet filtering is said to use implicit tagging.

In explicit tagging, the switch needs to know whether the device to which a packet is being sent is a VLAN aware device or not; in other words, whether the device is provided with the intelligence to interpret the tag information. The identifier is added only if the device is VLAN aware.

## 21.6 VLAN-Implementation methods

The membership of an end user device in a VLAN can be implemented in several ways.

In general, they will fall under one of the following five categories:

- Grouping by switch port
- Grouping by MAC address
- Grouping by network layer information
- Multicast group
- Higher layer grouping

We will review each of the approaches in detail.

### Grouping by switch port

In this method, the membership to a VLAN depends on the switch port to which it is connected. Two types of grouping are possible.

In the first, grouping can be within a single switch, say ports 1, 3, 5 and 7 make up one group and 2, 4, 6 and 8 make up a second group. (Refer to figure 21.5).

In the other method, which will involve more than one switch, a VLAN can include specific ports of different switches. As an example: ports 1, 2, 3 and 4 of switch-1 and ports 1,2, 3 and 4 of switch-2 may together form VLAN-A and ports 5, 6, 7 and 8 of switch-1 and ports 5, 6, 7 and 8 of switch-2 form another VLAN-B. (Refer to figure 21.6).

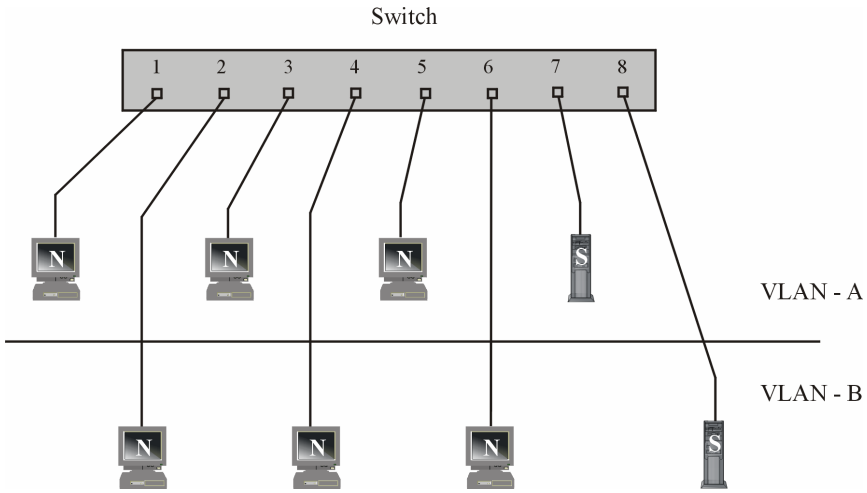



**Figure 21.5**
*Single switch with ports grouped*

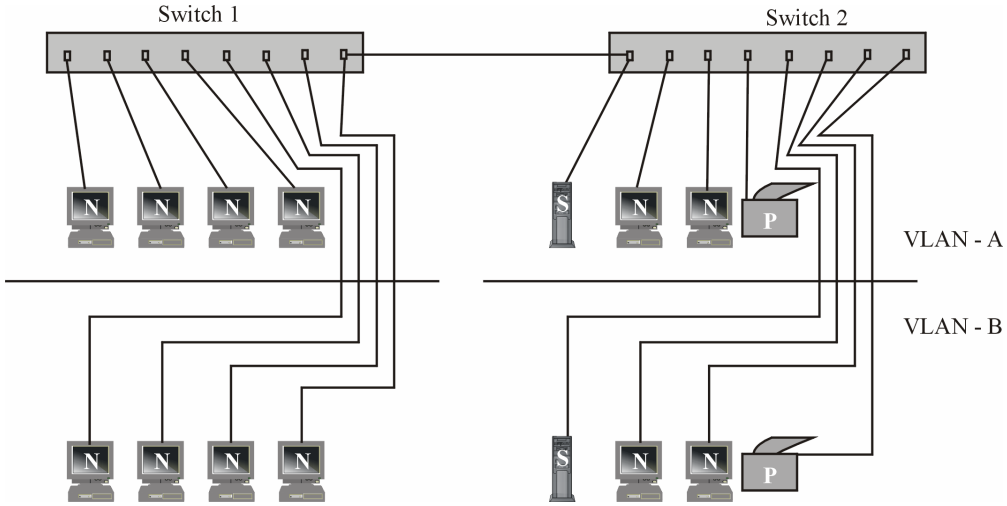



**Figure 21.6**
*Two switches with ports grouped*

The second method is the one that is more commonly adopted. The limitation of the method of using the switch port to indicate grouping is that whenever a particular user moves to a new location involving a different switch/port, the network administrator has to reconfigure the VLAN membership information.

Also, this grouping works best for nodes that are directly connected to a switch port. If a machine is connected to a port through a hub, all the machines connected to the hub have to be necessarily members of the same VLAN as one shared media or physical segment cannot be assigned to multiple VLANs. However, hubs with a multi-backplane connection feature overcome this limitation to some extent, since each backplane of the hub can be allocated to a particular VLAN.

The switch port grouped implementation works well in an environment where network moves are done in a controlled fashion and where robust VLAN management software is available to configure the ports.

### Grouping by MAC address

The problem of manual reconfiguration is overcome by grouping different machines based on their MAC Addresses. These addresses are machine specific and are decided based on the hardwired information unique to each individual network interface card (NIC). This ensures that whenever a user moves from one switch port to another, the grouping is still undisturbed and whatever is the move, the user's VLAN membership is retained without change. For this reason, this method can be thought of as a user based VLAN implementation.

This method has some drawbacks too. The MAC address based implementation requires that initially each MAC address be assigned to at least one VLAN. This means that to begin with, there will be one large VLAN containing a very large number of users and this may lead to some performance degradation. Some vendors who provide automated tools to create groups based on the initial network status overcome this. In other words, a MAC address based VLAN is created automatically for each subnet of the LAN.

Serious performance issues can arise if members of different VLANs coexist in a shared media environment (within a single segment). Also, in the case of large networks, the method of communicating the VLAN membership information between different switches to keep them always synchronized can cause performance degradation too.

The use of MAC address to identify VLAN membership can be problematic in a network where a large number of laptop computers are used and are connected to the network by means of docking stations. The NIC and therefore the MAC address is a part of the docking station, which usually remains on a particular desk. In a situation where the user and his/her laptop move around to different desks/docking stations their MAC address gets changed when they move to a different desk. This makes tracking groups based on MAC addresses difficult since reconfiguration will be needed whenever a user moves to a different docking station.

### Grouping by network layer information

This method is also called as Layer 3-based VLAN implementation. Here, grouping is based on layer 3 information such as protocol type, or in many cases network-layer address (such as subnet address in IP networks). The switches examine the information regarding the subnet address contained in each packet which decides to which VLAN the user belongs and make the routing decision on this basis. It should be noted that even though this method uses layer 3 information, this does not constitute routing. This is because no route calculation is done, frames are usually bridged according to the spanning tree algorithm and connectivity within any VLAN is seen as a flat-bridged topology. See figure 21.7 for an example of this type of implementation.

Some vendors, however, incorporate varying amounts of intelligence in their switches so that they carry out functions normally associated with layer 3 devices. Some of these layer 3 aware devices have the packet forwarding or routing function built into ASIC chips on the switch, which make them faster than CPU, based routers. However, the process of examining layer 3 addresses in a packet is definitely slower than looking at MAC addresses in frames. These implementations are therefore slower than those, which use layer 2 information.

VLANs grouped using layer 3 information are particularly effective in dealing with networks using TCP/IP than those, which use protocols such as IPX or DECnet, which do not involve manual configuration at the desktop. Particularly, end stations using un-routable protocols such as NetBIOS cannot be differentiated and therefore cannot be included in these VLANs.

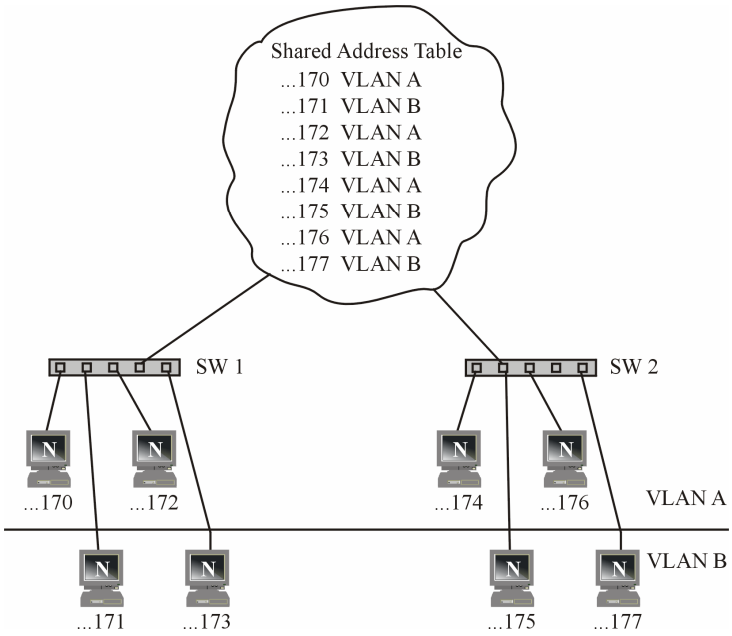



**Figure 21.7**
*VLAN grouping by network-layer address*

### VLANs using multicast groups

This represents a different approach to VLANs although the concept of a VLAN as a broadcast domain is still applicable. When an IP packet is sent via multicast, it is sent to an address that is a proxy for a group of IP addresses dynamically defined. Each workstation is given a chance to join this group for a certain period of time by sending an affirmative response to a broadcast. They are however members of the multicast group only for a certain period of time. This approach is thus highly flexible and exhibits a high degree of application sensitivity. Such groups can also span routers and therefore WAN connections.

### VLANs using higher layer grouping

VLANs implemented using this method of grouping are defined using applications or services or a combination. For example File Transfer Protocol (FTP) applications can belong to one VLAN and TELNET applications can belong to another. Such complex VLAN implementations need a high degree of automation in configuration and management.

### Combination of definitions

Considering that each of the above categories have their unique advantages and drawbacks, several vendors are attempting to use multiple methods of VLAN grouping in their products. This will give greater flexibility in configuring VLANs by network administrators depending on the specific features of their networks. For example a network may use both IP and NetBIOS protocols. The ability to define grouping based on IP Subnet addresses as well as by MAC addresses will enable separate VLANs to be set up in the same environment for IP subnets and NetBIOS end stations.

## 21.7 Method of connections

Devices in a VLAN can be connected in different ways depending on whether the device is ‘VLAN aware’ or ‘VLAN unaware’. As we saw earlier in this chapter, a VLAN aware

device can understand explicit tagging in frames and interpret to which VLAN the frame is to be directed.

These are called:

- Trunk link
- Access link and
- Hybrid link

We will discuss these in a little detail.

### Trunk link

All devices connected to a trunk link should be VLAN aware. This includes workstations. All frames on a trunk line should be explicitly tagged. These special frames are known as 'tagged frames'. Figure 21.8 below shows an example.

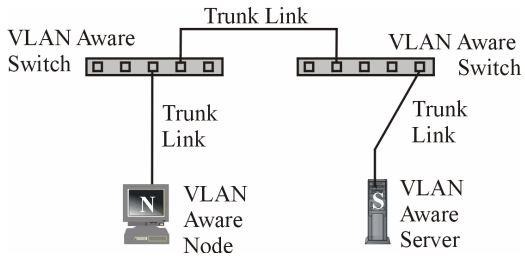



**Figure 21.8**
*An example of trunk link*

### Access link

An access link connects a VLAN unaware device to a VLAN aware bridge. It is usual to find VLAN unaware LAN segments in legacy LANs. Since VLAN unaware devices cannot handle explicitly tagged frames, the frames must be implicitly tagged, or in other words, they must be untagged. Refer to figure 21.9 for an illustration.

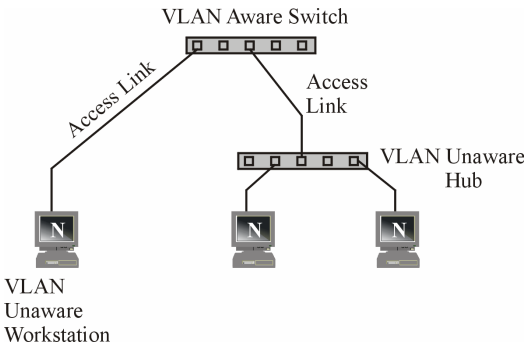



**Figure 21.9**
*Access link*

### Hybrid link

As the name indicates, Hybrid link is one that connects both VLAN aware and VLAN unaware devices together. A hybrid link can have tagged as well as untagged frames, but all the frames for a specific VLAN should be either tagged or untagged. Refer to figure 21.10 below.

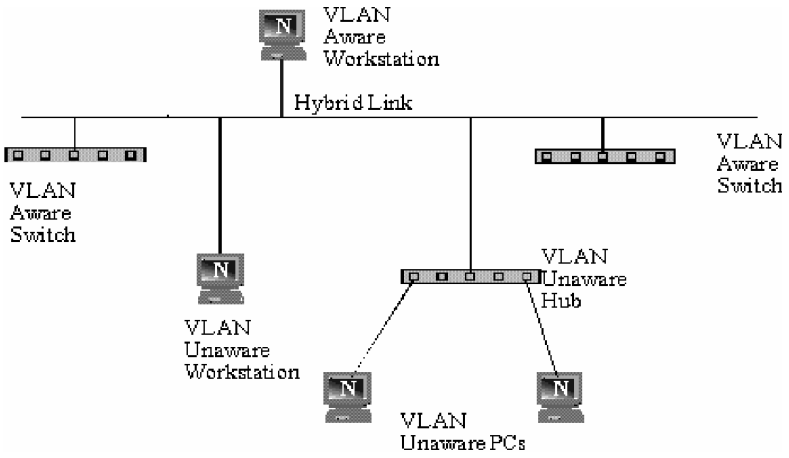



**Figure 21.10**
*Hybrid link*

All three types of links can be present in a single network.

## 21.8 Filtering table

As we saw earlier in this chapter, the filtering table in each switch is a database that stores the VLAN membership information that permits a switch to decide about how a packet is to be handled. There are essentially two types of entries, viz. static and dynamic.

### Static entries

The static entries are those that are created, modified or deleted by the network administrator. These are not automatically deleted by ageing, unlike dynamic entries.

Static entries can either be filtering entries or registration entries. A filtering entry is made for each port whether the frames to be specific address in a VLAN should be forwarded, discarded or should be decided by a dynamic entry. Static registration entries specify whether frames to be sent to a specific VLAN should be explicitly tagged and which ports are members of a VLAN.

### Dynamic entries

Dynamic entries are automatically created by the switch and cannot be added, or changed by the administrator. Unlike static entries they are automatically removed from the table after a certain time decided by he administrator and can vary between ten seconds to a million seconds. The learning process is by observing the packets sent by a port identifying the address and VLAN membership and the entries are made accordingly.

The entries can be of the dynamic filtering type, dynamic registration type or group registration type. Dynamic filtration entries specify whether a packet to be sent to a specific address in a particular VLAN should be forwarded or discarded. Dynamic registration entries specify as to which ports are registered in a specific VLAN. Addition or deletion of entries is done using a protocol called GVRP. Group registration entries indicate for each port whether frames to be sent to a group MAC address should be filtered or discarded. These entries are added or deleted using *Group Multicast Registration Protocol* (GMRP).

GVRP has another function too, that of communicating information to other VLAN aware switches. Such switches register and propagate VLAN membership to all the ports that are part of the currently active topology of the VLAN.

## 21.9 Tagging

Tagging is the process of adding information to the frame in the form of a tag header to enable switches to forward frames only to specific ports that belong to a particular VLAN instead of to all ports as done otherwise. Frames containing tag headers are called tagged frames.

The tag header also includes the following features:

- It allows user priority information to be specified
- It allows source routing information to be specified
- It includes MAC address format

The formats of tag frames are different for Ethernet and for Token Ring/FDDI networks. There are two parts to a tag header. They are: *Tag Protocol IDentifier* (TPID) and *Tag Control Information* (TCI). TPID is 2 bytes long for Ethernet and 8 bytes for Token Ring/FDDI, and contains a predefined number that identifies the frame as a modified (i.e. tagged) frame. TCI is 2 bytes long and contains a three-bit priority number as well as a twelve-bit VLAN ID number.

## 21.10 Summary

VLANs are useful when a network suffers from heavy broadcast traffic that needs to be contained or where the cost of frequent moves of network end user devices both due to recabling and due to consequent administrative overheads has become too high. They improve network security by preventing network traffic from going beyond the group for whom they are meant. It should be remembered that VLANs are proprietary solutions, which may need all hardware and management tools to be purchased from a single vendor. Also VLANs will NOT be effective if the network has a large presence of computers with shared media access. The actual implementation strategy will depend on the needs and practices of the user organization.